Year II — Mistake & Methods Log

Based on lectures by Various Lecturers

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Chapter 1

Mathematical Methods II

1.1 Line Integrals

1.1.1 Mistakes and Tips

Scalar Field vs Vector Field

A scalar field assigns a scalar value to each point in space, while a vector field assigns a vector to each point. For example, temperature can be modelled by a scalar field, whereas fluid velocity can be modelled by a vector field.

Scalar Line Integral vs Vector Line Integral

A scalar line integral accumulates a scalar field along a curve, weighted by arc length:

$$\int_C f \, d\ell.$$

If $\mathbf{x}(t)$ parametrises the curve for $t \in [a, b]$, then

$$\int_C f \, d\ell = \int_a^b f(\mathbf{x}(t)) \left| \frac{d\mathbf{x}}{dt} \right| dt.$$

For example, if f is linear mass density, this integral gives the mass of a wire.

A vector line integral (also called a work integral or circulation integral) measures the component of a vector field along the direction of travel:

$$\int_C \mathbf{F} \cdot \hat{\mathbf{t}} \, d\ell = \int_a^b \mathbf{F}(\mathbf{x}(t)) \cdot \frac{d\mathbf{x}}{dt} \, dt.$$

Unlike a scalar line integral, a vector line integral depends on the orientation of the curve: reversing the direction reverses its sign.

What does $f(\mathbf{x}(t))$ mean?

The parametrisation $\mathbf{x}(t) = (x(t), y(t), z(t))$ gives the point on the curve at parameter value t. To evaluate a scalar field $f(x, y, z)$ along the curve, substitute $x = x(t)$, $y = y(t)$ and $z = z(t)$, giving $f(\mathbf{x}(t)) = f(x(t), y(t), z(t))$. For a vector field, substitute in the same way into each component.

The speed $|\mathbf{x}'(t)|$ may vary

The quantity

$$|\mathbf{x}'(t)| = \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}$$

is the speed along the parametrised curve. It is generally a function of t, not a constant.

Parametrising implicit curves

If a curve is not already parametrised, choose a parameter that makes its equations and constraints easy to express. For circular or cylindrical symmetry, consider polar or cylindrical coordinates. Sketching the curve often helps identify the parameter range.

Splitting piecewise-smooth curves

If a curve consists of multiple smooth pieces, evaluate the integral on each piece and add the results. For vector line integrals, make sure each piece is parametrised in the direction of travel.

Unit tangent vectors and orthonormal bases

A unit tangent vector has magnitude 1. Unit tangent vectors associated with different coordinate directions are not automatically mutually orthogonal. Check their dot products before treating them as an orthonormal basis.

Length of a curve

If the question asks for the length of a curve, use the arc-length formula

$$L = \int_a^b |\mathbf{x}'(t)|\, dt = \int_{t_0}^{t_1} \sqrt{\frac{d\mathbf{x}}{dt} \cdot \frac{d\mathbf{x}}{dt}}\, dt.$$

Orientation matters for vector line integrals

Reversing the direction of travel changes the sign of $\int_C \mathbf{F} \cdot d\mathbf{x}$, but does not change a scalar line integral $\int_C f\, d\ell$.

1.1.2 Methods

Method 1.1.1 (Evaluating a Scalar Line Integral)**.**

(i) Parametrise the curve as $\mathbf{x}(t) = (x(t), y(t), z(t))$, for $a \le t \le b$.

(ii) Differentiate to find $\mathbf{x}'(t)$.

(iii) Calculate the speed $|\mathbf{x}'(t)|$.

(iv) Substitute the parametrisation into the field to obtain $f(\mathbf{x}(t))$.

(v) Evaluate

$$\int_C f\, d\ell = \int_a^b f(\mathbf{x}(t))|\mathbf{x}'(t)|\, dt.$$

Method 1.1.2 (Evaluating Arc Length).

(i) Parametrise the curve as $\mathbf{x}(t)$, for $a \leq t \leq b$.

(ii) Differentiate to find $\mathbf{x}'(t)$.

(iii) Calculate $|\mathbf{x}'(t)| = \sqrt{\mathbf{x}'(t) \cdot \mathbf{x}'(t)}$.

(iv) Integrate

$$L = \int_a^b |\mathbf{x}'(t)|\, dt.$$

Method 1.1.3 (Evaluating a Vector Line Integral).

(i) Parametrise the curve in the required direction: $\mathbf{x}(t)$, $a \leq t \leq b$.

(ii) Substitute into the vector field to obtain $\mathbf{F}(\mathbf{x}(t))$.

(iii) Differentiate the curve to find $\mathbf{x}'(t)$.

(iv) Calculate the dot product $\mathbf{F}(\mathbf{x}(t)) \cdot \mathbf{x}'(t)$.

(v) Evaluate

$$\int_C \mathbf{F} \cdot d\mathbf{x} = \int_a^b \mathbf{F}(\mathbf{x}(t)) \cdot \mathbf{x}'(t)\, dt.$$

Do not multiply by $|\mathbf{x}'(t)|$ in this formula: the derivative vector already accounts for direction and parameter speed.

Method 1.1.4 (Finding Unit Tangent Vectors in Orthogonal Coordinates). For a coordinate map $\mathbf{x}(u, v, w)$, define the scale factors

$$h_u = \left|\frac{\partial \mathbf{x}}{\partial u}\right|, \quad h_v = \left|\frac{\partial \mathbf{x}}{\partial v}\right|, \quad h_w = \left|\frac{\partial \mathbf{x}}{\partial w}\right|.$$

The corresponding unit coordinate tangent vectors are

$$\mathbf{e}_u = \frac{1}{h_u}\frac{\partial \mathbf{x}}{\partial u}, \quad \mathbf{e}_v = \frac{1}{h_v}\frac{\partial \mathbf{x}}{\partial v}, \quad \mathbf{e}_w = \frac{1}{h_w}\frac{\partial \mathbf{x}}{\partial w}.$$

These formulae apply where the scale factors are nonzero. The coordinate directions are orthogonal only if their tangent vectors are mutually perpendicular.

Method 1.1.5 (Checking an Orthonormal Basis). Check that each vector has unit magnitude and that distinct vectors are mutually orthogonal:

$$|\mathbf{e}_i| = 1, \qquad \mathbf{e}_i \cdot \mathbf{e}_j = 0 \quad (i \neq j).$$

If these conditions hold, the vectors form an orthonormal set; in three dimensions, three such vectors form an orthonormal basis.

Method 1.1.6 (Finding Vector Components in an Orthogonal Coordinate System).

(i) Rewrite the Cartesian components and the position in the desired coordinates using the coordinate transformations.

(ii) Find the unit coordinate vectors $\mathbf{e}_u, \mathbf{e}_v, \mathbf{e}_w$.

(iii) Express the field in the orthonormal basis by taking dot products:

$$F_u = \mathbf{F} \cdot \mathbf{e}_u, \qquad F_v = \mathbf{F} \cdot \mathbf{e}_v, \qquad F_w = \mathbf{F} \cdot \mathbf{e}_w.$$

(iv) Write $\mathbf{F} = F_u \mathbf{e}_u + F_v \mathbf{e}_v + F_w \mathbf{e}_w$.

Remember that changing the coordinate expressions of x, y, z is not enough by itself: vector components must be resolved along the new unit coordinate vectors.

1.2 Surface and Volume Integrals

1.2.1 Mistakes and Tips

Scalar Surface Integral vs Vector Surface Integral

A scalar surface integral accumulates a scalar quantity over a surface. For example, if f represents surface mass density, then

$$\iint_S f \, dS$$

gives the total mass distributed over the surface.

A vector surface integral is often used to calculate the flux of a vector field through an oriented surface:

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS.$$

The dot product with the unit normal measures the component of the field passing through the surface, rather than the component running parallel to it.

Scalar Volume Integral vs Vector Volume Integral

A scalar volume integral accumulates a scalar quantity throughout a three-dimensional region. For example, if f is a mass density, then

$$\iiint_V f \, dV$$

gives the total mass within the volume.

A vector volume integral integrates each component of a vector field separately. Its result is a vector, not a scalar:

$$\iiint_V \mathbf{F} \, dV = \begin{pmatrix} \iiint_V F_x \, dV \\ \iiint_V F_y \, dV \\ \iiint_V F_z \, dV \end{pmatrix}.$$

This is not flux through the boundary; flux is a surface integral.

Surface Area vs Scalar Surface Integral

Surface area is the special case of a scalar surface integral where the integrand is 1:

$$\text{Area}(S) = \iint_S 1 \, dS.$$

If $\mathbf{x}(u, v)$ parametrises S, then

$$\text{Area}(S) = \iint_U \left| \frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right| du \, dv.$$

For a general scalar field f, use

$$\iint_S f\,dS = \iint_U f(\mathbf{x}(u,v)) \left|\frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v}\right| du\,dv.$$

The notation Area(S) should only be used when the integrand is 1.

Volume vs Scalar Volume Integral

The volume of a region is the special case of a scalar volume integral where the integrand is 1:

$$\text{Vol}(V) = \iiint_V 1\,dV.$$

Under a coordinate transformation $\mathbf{x} = \mathbf{x}(u,v,w)$,

$$\text{Vol}(V) = \iiint_{V'} |J|\,du\,dv\,dw,$$

whereas a general scalar volume integral is

$$\iiint_V f(\mathbf{x})\,dV = \iiint_{V'} f(\mathbf{x}(u,v,w))|J|\,du\,dv\,dw.$$

In Cartesian coordinates, $|J| = 1$. The integrand is still $f(x,y,z)$ for a general scalar volume integral; use 1 only when calculating the volume itself.

Conceptual Understanding of Flux

Flux measures the net amount of a vector field passing through an oriented surface:

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}}\,dS.$$

The dot product selects the component of $\mathbf{F}$ in the normal direction. A field tangent to the surface contributes zero flux at that point. Reversing the normal reverses the sign of the flux.

The 3D Jacobian

For a coordinate transformation $\mathbf{x} = \mathbf{x}(u,v,w)$,

$$J = \frac{\partial(x,y,z)}{\partial(u,v,w)} = \det\begin{pmatrix} x_u & x_v & x_w \\ y_u & y_v & y_w \\ z_u & z_v & z_w \end{pmatrix} = \mathbf{x}_u \cdot (\mathbf{x}_v \times \mathbf{x}_w).$$

The absolute value $|J|$ is the local volume-scaling factor:

$$dV = |J|\,du\,dv\,dw.$$

In Cartesian coordinates the Jacobian determinant is 1. In a general transformation it need not be 1.

Do not always parametrise the whole surface at once

A surface such as a cube is piecewise smooth. Split it into its faces, calculate the contribution from each face, and add the results. For flux, ensure each face's normal has the required orientation.

Distinguish the normal vector from its magnitude

For a parametrised surface, $\mathbf{x}_u \times \mathbf{x}_v$ is a normal area vector. Its magnitude gives the surface-area scaling, while its direction gives the orientation. Do not replace this vector by its magnitude in a flux integral.

Number of parameters

A surface is generally parametrised by two parameters, while a three-dimensional volume is generally described using three coordinates or parameters.

Choose coordinates to simplify the region

Use Cartesian coordinates for regions bounded simply by planes or coordinate-aligned surfaces; cylindrical coordinates for circular symmetry or expressions involving x^2+y^2; and spherical coordinates for spheres and radial symmetry. Always sketch the region and check the bounds after changing coordinates.

Standard Volume Elements

- **Cartesian:**
$$dV = dx\,dy\,dz.$$
- **Cylindrical:** $x = r\cos\theta$, $y = r\sin\theta$,
$$dV = r\,dr\,d\theta\,dz.$$
- **Spherical:** using θ for the angle from the positive z-axis and ϕ for azimuth,
$$dV = r^2\sin\theta\,dr\,d\theta\,d\phi.$$

Check your lecture's spherical-coordinate convention, since some courses interchange θ and ϕ.

1.2.2 Methods

Method 1.2.1 (Systematically Finding Limits for a Volume Integral)**.**

(i) Sketch the region and identify its bounding surfaces.

(ii) Choose an order of integration, for example $dz\,dy\,dx$.

(iii) Find the lower and upper bounds for the innermost variable.

(iv) Project the region onto the plane of the two remaining variables and determine their bounds.

(v) Translate the bounds into the chosen coordinate system.

(vi) If one set of bounds cannot describe the region, split it into simpler regions and add the integrals.

Method 1.2.2 (Evaluating Surface Area)**.**

(i) Parametrise the surface as $\mathbf{x}(u, v)$ over a parameter domain U.

(ii) Calculate $\mathbf{x}_u$ and $\mathbf{x}_v$.

(iii) Calculate the magnitude $|\mathbf{x}_u \times \mathbf{x}_v|$.

(iv) Evaluate

$$\text{Area}(S) = \iint_U |\mathbf{x}_u \times \mathbf{x}_v| \, du \, dv.$$

(v) For a piecewise-smooth surface, calculate the area of each piece and add them.

Method 1.2.3 (Evaluating a Scalar Surface Integral)**.**

(i) Sketch or identify the surface and its parameter domain.

(ii) Parametrise the surface as $\mathbf{x}(u, v)$, or split a piecewise-smooth surface into faces.

(iii) Calculate $\mathbf{x}_u$, $\mathbf{x}_v$ and $|\mathbf{x}_u \times \mathbf{x}_v|$.

(iv) Substitute the parametrisation into the scalar field and evaluate

$$\iint_S f \, dS = \iint_U f(\mathbf{x}(u, v)) |\mathbf{x}_u \times \mathbf{x}_v| \, du \, dv.$$

(v) If the surface has multiple pieces, add the integral over each piece.

Method 1.2.4 (Evaluating Flux Through a Parametrised Surface)**.**

(i) Parametrise the surface as $\mathbf{x}(u, v)$ over U.

(ii) Calculate $\mathbf{x}_u \times \mathbf{x}_v$.

(iii) Check the direction of this normal area vector. If it points opposite to the required orientation, reverse its sign.

(iv) Substitute $\mathbf{x}(u, v)$ into the vector field $\mathbf{F}$.

(v) Take the dot product with the *vector* normal area element and integrate:

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS = \iint_U \mathbf{F}(\mathbf{x}(u, v)) \cdot (\mathbf{x}_u \times \mathbf{x}_v) \, du \, dv.$$

(vi) For a piecewise-smooth surface, sum the flux through all pieces using the required orientation on each.

Method 1.2.5 (Surface Given as a Graph $z = g(x, y)$)**.** For the upward orientation, parametrise the graph by $\mathbf{x}(x, y) = (x, y, g(x, y))$. Then

$$\mathbf{x}_x \times \mathbf{x}_y = (-g_x, -g_y, 1).$$

Hence, for a scalar field,

$$\iint_S f \, dS = \iint_D f(x, y, g(x, y)) \sqrt{1 + g_x^2 + g_y^2} \, dx \, dy,$$

and for flux,

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}}\, dS = \iint_D \mathbf{F}(x, y, g(x, y)) \cdot (-g_x, -g_y, 1)\, dx\, dy.$$

For downward orientation, reverse the sign of the normal vector in the flux formula.

Method 1.2.6 (Finding the Volume of a Region)**.**

(i) Sketch the region and determine the integration limits.

(ii) Choose Cartesian, cylindrical, spherical, or another suitable coordinate system.

(iii) Use the corresponding volume element, including its Jacobian:

$$\mathrm{Vol}(V) = \iiint_{V'} |J|\, du\, dv\, dw.$$

(iv) Integrate over the transformed region V'.

Method 1.2.7 (Evaluating a Scalar Volume Integral)**.**

(i) Sketch the region and determine the limits of integration.

(ii) Choose coordinates that simplify the integrand or the region.

(iii) Rewrite the field in the new coordinates and calculate the absolute Jacobian $|J|$.

(iv) Evaluate

$$\iiint_V f(x, y, z)\, dV = \iiint_{V'} f(\mathbf{x}(u, v, w))|J|\, du\, dv\, dw.$$

(v) In Cartesian coordinates, $|J| = 1$; do not replace f by 1 unless you are calculating the volume itself.

Method 1.2.8 (Evaluating a Vector Volume Integral)**.** If $\mathbf{F} = (F_x, F_y, F_z)$, integrate each component separately:

$$\iiint_V \mathbf{F}\, dV = \begin{pmatrix} \iiint_V F_x\, dV \\ \iiint_V F_y\, dV \\ \iiint_V F_z\, dV \end{pmatrix}.$$

Use the same region, limits and volume element for all three components. Under a coordinate transformation, include the absolute Jacobian $|J|$ in each component integral.

Chapter 2

Algebra II

2.1 Rings, Fields and Subrings

2.1.1 Mistakes and Tips

Determining unknown binary operations

If not specified, assume that R is a ring with $+$ and $\cdot$ by default

Using universal conditions

Work with any universal condition when manipulating or showing structure and then use strategically chosen elements

Strategically choosing elements

Use strategically chosen elements that exploit certain conditions such as $a+b$ or ab

Dividing in rings

Never divide by elements unless you know it is invertible. This is because rings do not require multiplicative inverses. Instead use addition / subtraction on either side and also distributivity.

Make use of earlier results

Utilise earlier results of the question where possible.

Simplifying expansions

Do not simplify expansions with hypotheses until it is the direct next step as you lose the algebra needed to show it is true.

Utilising LHS and RHS

Showing properties like associativity, commutativity and distributivity can be done easily by showing LHS = RHS.

Efficiently forming Modular Arithmetic tables

When dealing with integers modulo n, it is smart to use a multiplication and addition table. To quickly fill them in, consider how much over a multiple of n your entry to the table is to find the residue.

Using fields for subrings

Using the fact that a subring of a ring is a ring itself allows you to quickly prove rings if you know that they are a subring of a larger field like $\mathbb{C}$.

Dealing with subring questions

When dealing with a subring question, make sure to use inherited elements and operations to show certain structures. For example 0 and 1.

Showing subrings are the parent rings

Showing subrings are the ring themselves requires showing $R \subset S$ as $S \subset R$ by default.

Using counterexamples for disproofs

Finding one failure allows you to disprove closure and the full ring quite easily. Use counterexamples strategically here.

2.1.2 Methods

Method 2.1.1 (Determining a Ring)**.** To determine a ring, one should first check that the ring forms an abelian group with respect to the 'addition' operation and that the following requirements hold for the 'multiplication' operation:

i) **Existence of an Identity** - There exists an element $1 \in R$ such that $1 \cdot x = x \cdot 1 = x$, for all $x \in R$.

ii) **Associativity** - For any $x, y, z \in R$ we have $x(yz) = (xy)z$.

iii) **Distributivity** - For any $x, y, z \in R$ we have $x(y + z) = xy + xz$ and $(y + z)x = yx + zx$.

Method 2.1.2 (Showing a Subring)**.** We can quickly show a subring by showing:

- the additive and multiplicative identities exist in the set,
- it's closed under addition
- it's closed under multiplication
- it contains additive inverses.

Method 2.1.3 (Determining a Ring quickly)**.** Rather than displaying a set satisfies all of the properties of a ring, it is sufficient to show that the set is a subring of a ring. It is therefore a ring itself.

Method 2.1.4 (Showing commutativity)**.** Showing commutativity requires introducing arbitrary elements and comparing expressions such as $(a+b)^2$ and the additive inverses to show equivalence.

Method 2.1.5 (Quick way to see that a set is not a ring)**.** If multiplication is not closed by default then a set fails to be a ring as it is a requirement of the binary operation.

Method 2.1.6 (Disproving $\mathbb{Z}/n$ as a field)**.** Showing that $\mathbb{Z}/n$ is not a field requires finding 2 single none zero elements that are zero divisors such that:

$$ab = 0 \bmod(n).$$

This immediately disproves a field. Check columns of a table here to do this quickly.